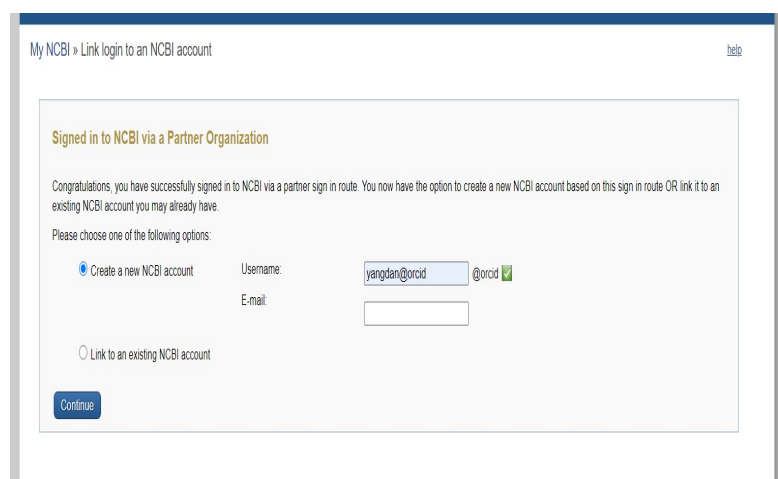
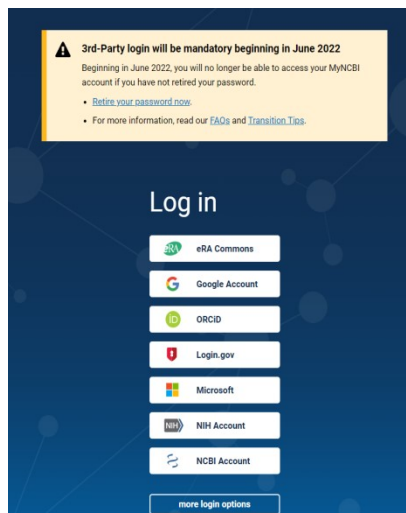
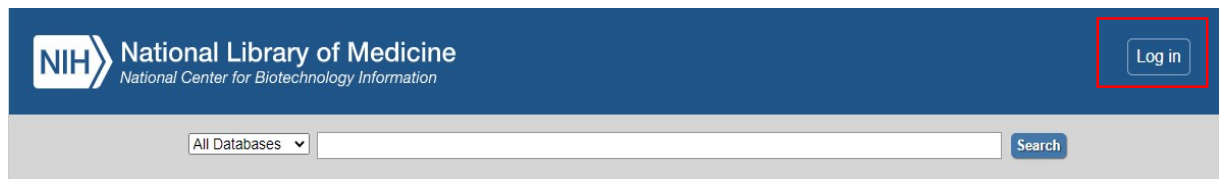


SRA 数据上传操作指南

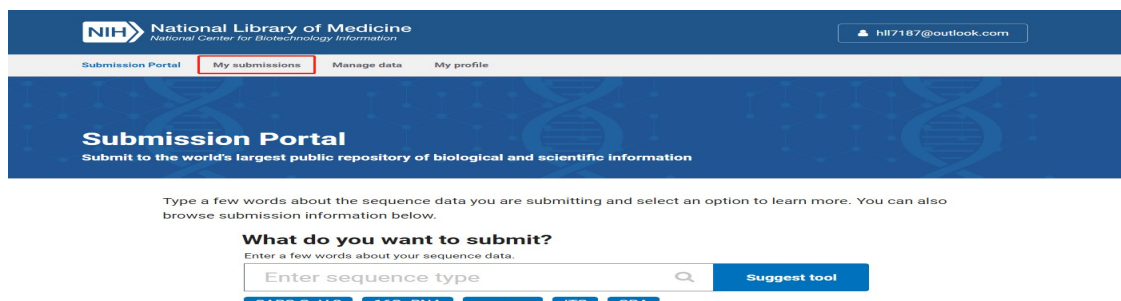
一、SRA 数据上传操作方法

1、NCBI 账号注册

数据上传前需要注册一个 NCBI (<https://www.ncbi.nlm.nih.gov/>) 账号, 创建需要邮件激活。如果已经有账号, 可以直接点击登录。在 NCBI 的首页点 Submit, 选择 Sequence Read Archive (SRA), 点击 GO, 点击 New submission。



2、Bioproject 创建：用于描述对样本进行测序的研究目的

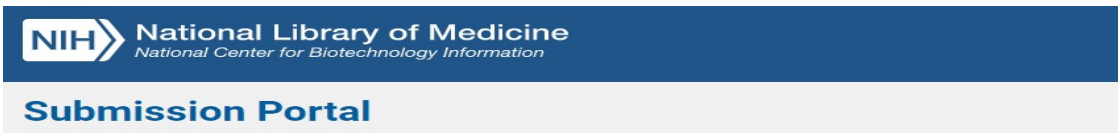


Your submissions

Start a new submission

- GenBank
- Sequence Read Archive
- Genome
- TSA
- BioProject
- BioSample
- Supplementary Files
- API

You have not submitted anything yet.



BioProject

[New submission](#)

点击



ATTN: to update an existing record, you can use "Manage data" to make some changes yourself. If you cannot make the desired change there, then please email your request with your BioProject accession or Submission ID included.

A BioProject is a collection of biological data related to a single initiative, originating from a single organization or from a consortium of coordinating organizations.

BioProjects aggregate pointers to data to provide users with an entry point into diverse data types.



Info: You have not submitted anything yet in this new BioProject registration system.

填写 submitter 详细信息：带“*”号为必填（姓名、邮箱、单位、地址、邮政编码），点击 continue。

BioProject submission: SUB4409411

New

[Delete submission](#)

1 SUBMITTER 2 PROJECT TYPE 3 TARGET 4 GENERAL INFO 5 BIOSAMPLE 6 PUBLICATIONS 7 OVERVIEW

Submitter

Required fields are marked with * asterisk

* First (given) name Middle name * Last (family) name

* E-mail (primary) E-mail (secondary) At least one e-mail should be from the organization's domain.

Group for this submission:
[No group](#) (affiliation from my personal profile)

[Create group for shared submissions](#) Allow selected collaborators to read, modify, submit and delete your submissions

* Submitline organization Submitline organization ID# * Department

BioProject submission: SUB4409411

New

[Delete submission](#)

1 SUBMITTER 2 PROJECT TYPE 3 TARGET 4 GENERAL INFO 5 BIOSAMPLE 6 PUBLICATIONS 7 OVERVIEW

Project Type

Required fields are marked with * asterisk

* Project data type

- ☐ Genome sequencing and assembly
- ☐ Raw sequence reads
- ☐ Genome sequencing
- ☐ Assembly
- ☐ Clone ends
- ☐ Epigenomics
- ☐ Exome
- ☐ Map
- ☐ Metagenome
- ☐ Metagenomic assembly
- ☐ Phenotype or Genotype
- ☐ Proteome
- ☐ Random survey
- ☐ Targeted loci cultured
- ☐ Targeted loci environmental
- ☐ Targeted Locus (Loci)

项目类型基因组/转录组，外泌体，宏基因组等

BioProject submission: SUB4409411

Raw sequence reads

[Delete submission](#)

1 SUBMITTER 2 PROJECT TYPE 3 TARGET 4 GENERAL INFO 5 BIOSAMPLE 6 PUBLICATIONS 7 OVERVIEW

Target

The most descriptive organism name for the study (to the species, if relevant).

研究物种信息

Required fields are marked with * asterisk

* Organism name

Strain Breed Cultivar Isolate name Label

Description

Continue

1 SUBMITTER 2 PROJECT TYPE 3 TARGET 4 GENERAL INFO 5 BIOSAMPLE 6 PUBLICATIONS 7 OVERVIEW

General Info

Required fields are marked with * asterisk

Release date

Note: Release of BioProject or BioSample is also triggered by the release of linked data.

When should this submission be released to the public?

Release immediately following processing

Release on specified date or upon publication, whichever is first

* Projected release date

* Project title

Aedes novoniveus Raw sequence reads

* Public description

Relevance

信息释放时间节点, 项目的主要概况, 研究目的等

Submission Portal

Home My submissions Templates My profile

BioProject submission: SUB4409411

Aedes novoniveus Raw sequence reads

[Delete submission](#)

1 SUBMITTER 2 PROJECT TYPE 3 TARGET 4 GENERAL INFO 5 BIOSAMPLE 6 PUBLICATIONS 7 OVERVIEW

BioSample

Required fields are marked with * asterisk

Sample

If you have not registered your sample, please [register at BioSample](#). At the end of that process, you will be returned to this submission.

Please note that only single biosamples can be registered via this link. To register multiple/batch biosamples, complete your bioproject without registering biosamples and then submit the biosamples separately, including the bioproject accession in the submission.

Click 'Continue' without selecting a BioSample to skip this step. Note that links can be made after a BioSample is registered separately.

如果已经创建 Biosample, 则填写 BioSample 编号, 否则创建 BioSample

Continue

BioProject submission: SUB4409411

Aedes novoniveus Raw sequence reads

[Delete submission](#)

1 SUBMITTER 2 PROJECT TYPE 3 TARGET 4 GENERAL INFO 5 BIOSAMPLE 6 PUBLICATIONS 7 OVERVIEW

Publications

Required fields are marked with * asterisk

PubMed ID DOI

Add another publication

添加完 Biosample 后进入 Publication 界面, 如果有发表的文章则填写文章的 Pubmed ID/DOI 号

Continue

National Center for Biotechnology Information

8600 Rockville Pike
Bethesda MD, 20894 USA
About Us Contact Us Policies ENA

NCBI Support Center

U.S. National Library of Medicine
National Institutes of Health
Department of Health and Human Services
USA.gov

BioProject submission: SUB4409411

Aedes novoniveus Raw sequence reads

[Delete submission](#)

1 SUBMITTER 2 PROJECT TYPE 3 TARGET 4 GENERAL INFO 5 BIOSAMPLE 6 PUBLICATIONS 7 OVERVIEW

Overview

This BioProject submission will be released **immediately following processing**.

Submitter

Submitter Wang Baojian
wangbaojian@novogene.com

Submitting organization Novogene

Project type

Sample scope Monoisolate

BioSample SUB4409421

Target

To make any necessary changes before submitting click on the tabs/steps above.

To proceed please review your submission, make changes, then click Submit button below.

Submission	Title	Status	Updated
SUB11567081	Actinobacillus hominis Transcriptome	✓ BioProject: Processed PRJNA846532 : Actinobacillus hominis Transcriptome or Gene expression (TaxID: 719) Manage data	Jun 07

3、Biosample 创建:用于描述测序样本的具体信息

The screenshot shows the NCBI homepage. In the left-hand navigation menu, under 'All Databases', 'BioSample' is highlighted. The main content area shows the NCBI logo and a brief description of the National Center for Biotechnology Information.

The screenshot shows the BioSample database landing page. Red annotations highlight key features:

- A red box around 'BioSample Overview' in the 'Using BioSample' section, with a red arrow pointing to it and the text '该页面详细介绍了 BioSample 的用途, 注意事项以及视频教程' (This page introduces the use of BioSample, precautions, and video tutorials).
- A red box around the 'Submit' link in the 'Search Help' section, with a red arrow pointing to it and the text '已经有该项目的 bioproject 直接点击 submit' (There is already a bioproject for this project, click submit directly).

The screenshot shows the BioSample submission portal. Red annotations highlight key features:

- A red arrow pointing to the 'New submission' button with the text '← 点击新建' (Click to create new).
- A red box around the 'Submitter' section, with the text '填写基本信息' (Fill in basic information).

Submission Portal

BioSample [New submission](#) ← 点击新建

Submitter

★ First (given) name: Yefeng Middle name: Last (family) name: Xia

★ Email (primary): 704937611@qq.com ★ Email (secondary): xiaiefeng@novogene.com

Group for this submission: No group (affiliation from my personal profile)

★ Submitting organization: Homo Submitting organization URL: Department: Novogene

★ Street: Wuqing ★ City: Tianjin ★ State/Province: ★ Postal code: 300000 ★ Country: China

General Information

Release date

Note: Release of BioProject or BioSample is also triggered by the release of linked data.

★ When should this submission be released to the public?

- ☐ Release immediately following processing
- ☒ Release on specified date or upon publication, whichever is first

选择信息何时向公众公开

★ Projected release date

2021-11-30

★ Specify if you are submitting a single sample or a file containing multiple samples

- ☒ Batch/Multiple BioSamples
You will be asked to upload a tab-delimited text file that describes each of your samples and their attributes. Submission template files can be downloaded from the Attributes tab or the [templates page](#).
- ☐ Single BioSample
You will be asked to manually complete a web form to describe one sample and its attributes.

Sample Type

★ Select the package that best describes your samples.

All packages Packages for MAG submitters Packages for metagenome submitters

添加物种信息

(Optional) Filter packages by organism name

Enter the full scientific name of your samples, e.g., *Escherichia coli*

Reset and show all packages

- To filter for relevant BioSample packages, enter the full scientific name of the organism of your samples.
- If your BioSamples are derived from a species not represented in NCBI's Taxonomy database, enter the genus-level name, e.g., *Escherichia*
 - If your BioSamples are derived from more than one organism, enter the common species, genus, or family, e.g., *Enterobacteriaceae*
 - If your BioSamples are metagenomic/environmental, or metagenome-assembled genomes (MAG), select the appropriate tab above

NCBI packages [More...](#)

- ☐ Pathogen
Use for pathogen samples that are relevant to public health. Required attributes include those considered useful for the rapid analysis and trace back of pathogens.
- ☐ Microbe
Use for bacteria or other unicellular microbes when it is not appropriate or advantageous to use [MxS](#), Pathogen or Virus packages.
- ☐ Model organism or animal
Use for multicellular samples or cell lines derived from common

GSC MxS packages for genomes, metagenomes, and marker sequences [More...](#)

- ☐ MIGS Cultured Bacterial/Archaeal
Use for cultured bacterial or archaeal genomic sequences. Organism must have lineage [Bacteria](#) or [Archaea](#).
- ☐ MIGS Eukaryotic
Use for eukaryotic genomic sequences. Organism must have lineage [Eukaryota](#).
- ☐ MIGS Viral
Use for virus genomic sequences. Organism must have lineage [Viruses](#).
- ☐ MIMAG Metagenome-assembled Genome

选择物种类型

Submission Portal

BioSample submission: SUB11940843

MIUVIG Uncultivated Virus Genome sample

Attributes

填写或下载填写样本相关表格

Package MIUVIG: uncultivated virus genome, human-gut; version 5.0

★ How do you want to provide your BioSample attributes?

- ☐ Use built-in table editor
- ☒ Upload a file using Excel or text format (tab-delimited) that includes the attributes for each of your BioSamples

将填写好的样本上传

★ Attributes file

[Choose file](#) or drag and drop it here

如果样本不多，推荐直接在网页填写表格

Template for BioSample package MIUVIG: uncultivated virus genome, human-gut; version 5.0

[Download Excel](#) or [download TSV](#)

For column explanations and examples, please see the [sample attributes page](#).

For more information, please see [creating sample attribute file](#).

Continue

网页版表格：必填字段标有*星号。标有**，††或‡‡的字段至少填写其中一个。

表格填写：不同的生物样品类型所需填写的表格内容不同。绿色字段为必填项，如缺少相关信息，如任何必填项没有资料，可以填写 'not collected', 'not applicable' 或 'missing'; 以人类样品为例，至少需要填写绿色单元格里的以下几项样品信息：

- (1) 样品名称 (sample name)：无特定格式要求。
- (2) 物种 (organism)：实验物种的拉丁名。
- (3) 个体 (isolate)：为该实验样品个体取一个名字，无特定格式要求。
- (4) 年龄 (age)：格式为【数字】+【时间单位】。
- (5) 单位 (biomaterial_provider)：实验室或 PI 的名称和地址等信息，无特定格式要求。

(6) 性别 (Sex)：请从下拉菜单中选取符合条件的选项，而不要手动填写。

(7) 组织 (tissue)：样品来源的组织类型，无特定格式要求。

蓝色字段为至少一个必填项，可参考绿色字段填写方法；黄色字段则为非必填项。注意，除样本名外，**需要可以有信息将每个样本区分开**，可以通过添加 description 信息，或者另外增加 replication 的信息来进行区分（注意，这里信息如果填写不符合规范，上传后会报错，只需根据提示的信息对表格内容进行相应修改后重新上传即可）

若多个样品填写为相同的信息（生物学重复），需要在最后一列添加“replicate”(replicate=replicate biological X(1、2、3))

4、数据上传导航

Sequence Read Archive (SRA) submission: SUB11580613

New

Delete submission

1 SUBMITTER

2 GENERAL INFO

3 SRA METADATA

4 FILES

5 REVIEW & SUBMIT

Submitter

Required fields are marked with *

★ First (given) name

dan

Middle name

★ Last (family) name

yang

★ Email (primary)

yangdan202206@126.com

Email (secondary)

At least one email should be from the organization's domain.

Group for this submission

No group (affiliation from my personal profile)

Create group

Allow selected collaborators to read, modify, submit and delete your submissions

1 SUBMITTER

2 GENERAL INFO

3 PROJECT INFO

4 SRA METADATA

5 FILES

6 OVERVIEW

General Information

Required fields are marked with * asterisk

BioProject

BioProject describes the goal of your research effort.

Did you already register a BioProject for this research, e.g. for the submission of the reads to SRA and/or of the genome to GenBank?

☒ Yes

☐ No

★ Existing BioProject

BioSample

The BioSample records the detailed biological and physical properties of the sample that was sequenced. A BioSample can be used in more than one BioProject since it should be used for all the data that were obtained from that sample. Usually SRA data sets are generated from more than one sample.

Did you already register a BioSample for this sample, e.g. for the submission of the reads to SRA and/or of the genome to GenBank?

☒ Yes

☐ No

Release date

Note: Release of BioProject or BioSample is also triggered by the release of linked data.

Submission Portal

Home

My submissions

Templates

My profile

Sequence Read Archive (SRA) submission: SUB4409433

New

Delete submission

1 SUBMITTER

2 GENERAL INFO

3 PROJECT INFO

4 SRA METADATA

5 FILES

6 OVERVIEW

Project Info

Required fields are marked with * asterisk

★ Project title

★ Public description

Relevance

★ Is your project part of a larger initiative which is already registered with NCBI?

☒ No

☐ Yes (not very common)

External Links

NIH

U.S. National Library of Medicine

National Center for Biotechnology Information

wangbaojian

Submission Portal

Home

My submissions

Templates

My profile

Sequence Read Archive (SRA) submission: SUB4409433

AAA, Aug 17 '18

Delete submission

1 SUBMITTER

2 GENERAL INFO

3 PROJECT INFO

4 SRA METADATA

5 FILES

6 OVERVIEW

SRA metadata

Required fields are marked with * asterisk

For more detailed help with SRA submission please read the [SRA Submission Wizard Help](#).

★ How do you want to provide your metadata?

☐ Use built-in table editor

☐ Upload a file using Excel or text format (tab-delimited)

Continue

填写 Bioproject 和 BioSample 号即可，如果没有填写 no 即可，在数据上传的后面步骤中 会让您补充 Bioproject 和 BioSample 信息。

项目基本描述信息

选择添加 metadata 的方式，是在线填写或者下载表格填写后上传

我们建议您勾选“Upload a file using Excel or text format (tab-delimited)”并点击下图红框内的“Download Excel spreadsheet”超链接下载模板表格文件，以便在


```
Microsoft Windows [版本 10.0.16299.371]
(c) 2017 Microsoft Corporation。保留所有权利。

C:\Users\lihzh>cd C:\Users\lihzh\AppData\Local\Programs\Aspera\Aspera Connect\bin

C:\Users\lihzh\AppData\Local\Programs\Aspera\Aspera Connect\bin>
C:\Users\lihzh\AppData\Local\Programs\Aspera\Aspera Connect\bin>
C:\Users\lihzh\AppData\Local\Programs\Aspera\Aspera Connect\bin>
C:\Users\lihzh\AppData\Local\Programs\Aspera\Aspera Connect\bin>
C:\Users\lihzh\AppData\Local\Programs\Aspera\Aspera Connect\bin>ascp -i %key_file% %upload_path%\result\%file_name%.rawdata subasp@upload.ncbi.nlm.nih.gov:uploads/2018-08-08-0800-qc-kfz-wxv7

5% 1137KB 4.4MB/s 02:22 ETA
```

The screenshot shows the 'Select preload folder' dialog box. The dialog has a title bar 'Select preload folder' with a close button. Below the title bar, there is a note: 'Please note: it takes at least 10 minutes for uploaded files to become available here.' and a 'Refresh folders' button. The main area displays a table of files for upload:

Name	Files	Size	Modified
Aspera Data	7	8.4 GB	2019-10-31 20:33
gz.md5 (1.9 kB)			
Shamrock_M045-01-T01_good_1.fq.tar.gz (1.3 GB)			
Shamrock_M045-01-T01_good_2.fq.tar.gz (1.3 GB)			
Shamrock_M045-01-T02_good_1.fq.tar.gz (1.5 GB)			
Shamrock_M045-01-T02_good_2.fq.tar.gz (1.6 GB)			
...			

At the bottom of the dialog, there are two buttons: 'Use selected folder' and 'Cancel'. The 'Select preload folder' button in the left sidebar is highlighted with a red box. An orange arrow points from the 'Use selected folder' button to the 'Select preload folder' button in the sidebar.

二、SRA 数据上传常见问题

Biosample 创建中 Attributes 表格填写, 需要注意除样本名外, 需要可以有信息将每个样本区分开, 可以通过添加 description 信息, 或者另外增加 replication 的信息来进行区分。有生物学重复的需要 在最后一列添加 “replicate”(replicate=replicate biological X(1、2、3))

可以降低网速重新尝试，或者检查上传文件路径是否有空格，以及文件命名是否有空格，检查后更改路径重新上传。

3、输入代码后 error:key passphrase:

退出软件，重新启动或重新下载

4、传输中断；error: 0bit/s session stop:

重新输入代码，并建议适当调低网速

5、Session Stop (Error: Failed to open TCP connection for SSH)

重新下载软件重新登录再尝试

总结：目前最常见的问题是表格上传问题可以重点检查上传表格中填写是否完全一致，无法区分样本。及输入代码后不弹出上传的命令问题可以检查网速，检查上传文件路径是否有空格，以及文件命名是否有空格，检查后更改路径重新上传。